Question 1

a)

Active power transfer capacity of a bipole HVDC link
$P_{DC} = 2\,V_{DC} I_{DC}$

Active power transfer capacity of three-phase AC line
$P_{AC} = 3\,V_{LN} I_{AC} \cos\phi$

Identical conductor rating for AC and DC option implies:
$I_{AC} = I_{DC}$

Identical insulator rating for AC and DC option implies:
$\sqrt{2} V_{LN} = V_{DC}$

Ratio of active power transfer capacity for three-phase AC and bipole HVDC is:
$$\frac{P_{AC}}{P_{DC}} = \frac{3\,V_{LN} I_{AC} \cos\phi}{2\,V_{DC} I_{DC}} = \frac{3\,V_{LN} I_{AC} \cos\phi}{2\sqrt{2} V_{LN} I_{AC}} \approx 1$$
assuming $\cos\phi = 0.9 - 0.95$

[1 mark for each point – 5 marks]

b)

- Low order harmonics on both AC and DC sides are mitigated by using a star-star and star-delta connection for the two three-phase transformers connected to the two 6-pulse converter bridges in each pole.

[1 mark]

- Star-delta connection creates a 30° phase difference between the line and phase quantities. This eliminates the 5th and 7th order current harmonics on the AC side by creating a stair-case like waveform out of rectangular current pulses for each phase.

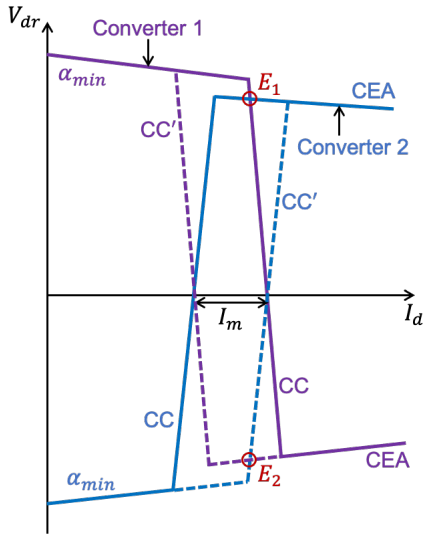
[2 marks]

- This also reduces the voltage harmonics on the DC side by creating 12 pulses out of the 6 pulses from individual 6-pulse converters. The odd multiples of 6th order harmonics are eliminated as the two 6-pulse bridge in series produce them out of phase.

[2 marks]

Model Answers

c)



- Intersection of the solid lines i.e., operating point E_1 depcits the situation when power transfer is from converter 1 to 2
- Dotted line characteristics i.e., operating point E_2 represents reverse power flow from converter 2 to 1
- Power flow reversal is achieved by making the current order setting (I_m) of converter 2 more than that of 1
- In practice, power reversal is done by swapping the current order setting between the two ends of the LCC-HVDC link

[3 marks for sketch and labelling, 2 marks for description]

d)

- As the direct voltage at rectifier end and hence direct current reduces, the rectifier tries to maintain direct current by reducing the gate delay angle (α)
- Depending on the extent of reduction in rectifier direct voltage, the rectifier could reach its mimimum gate delay angle limit (α_{min}) and switch to constant gate delay angle mode
- Inverter takes over current control
- Inverter regulates direct current at $(I_{ord} - I_m)$ by increasing extinction advance angle (γ) which leads to higher reactive power consumption
- For large reduction in direct voltage voltage-dependent current order limit kicks in to reduce the current/power order and continue the operation of the LCC-HVDC link with reduced power

[1 mark for each point – 5 marks]

Model Answers

e)

- Commutation failure is the failure to complete commutation of current from (say) thyristor 1 to 3 before reversal of commutating voltage (e_{ba})
- It happens when the outgoing thyristor (say, thyristor 1) fails to turn off before e_{ba} reverses polarity after which the current through the incoming thyristor (say, thyristor 1) is forced to go down to zero.
- Commutation failure is likely to occur when the direct current is high and/or the AC side voltage is low.
- If the current failts to transfer from thyristor 1 to 3, thyristor 1 continues to conduct when thyristor 4 comes into conduction creating a short circuit across the converter bridge.
- As a result, direct current increases drastically leading to successive commutation failure in subsequent cycles.

[1 mark for each point – 5 marks]

Model Answers

Question 2

a)

$$S_1 = \frac{V^2}{Z_1} = \frac{400^2}{3.5} = 45.71 \text{ GVA}$$
$$S_2 = \frac{V^2}{Z_2} = \frac{400^2}{5.5} = 29.09 \text{ GVA}$$

[1 mark]

$$\text{ESCR} = \frac{S - Q_c}{P_{DC}}$$

[1 mark]

$$Q_c = 0.6 \times P_{DC}$$

$$\Rightarrow P_{DC} \leq \frac{S}{\text{ESCR}+0.6}$$

[1 mark]

$$P_{DC-1} \leq \frac{S_1}{\text{ESCR}+0.6} = 9.43 \text{ GW}$$

$$P_{DC-2} \leq \frac{S_2}{\text{ESCR}+0.6} = 6.0 \text{ GW}$$

So maximum possible rating of the HVDC links is 6.0 GW

[2 marks]

b)

i)

No of 6-pulse bridge per pole $B = 2$

Rated current

$$I_d = \frac{P_d/2}{V_d} = 3.75 \text{ kA}$$

[1 mark]

Inverter end maintains the rated voltage $V_{di} = 800$ kV while operating at the minimum extinction angle by adjusting the effective turns ratio of the converter transformer

[1 mark]

Model Answers

$$V_{d0i} = \frac{V_{di} + BR_c I_d}{\cos(\gamma_{min})} = \frac{3\sqrt{2}}{\pi} BT_i E_{ac} \Rightarrow T_i = \frac{(V_{di} + BR_c I_d)\pi}{3\sqrt{2}BE_{ac}\cos(\gamma_{min})}$$

[2 marks]

With $E_{ac} = E_{ac-rated} = 400$ kV

$$T_i = \frac{(V_{di} + BR_c I_d)\pi}{3\sqrt{2}BE_{ac-rated}\cos(\gamma_{min})} = 0.8281$$

[1 mark]

Direct voltage at rectifier end

$V_{dr} = V_{di} + R_L I_d = 822.49$ kV

[1 mark]

Rectifier end maintains the rated current while operating at the nominal firing angle by adjusting the effective turns ratio of the converter transformer

[1 mark]

$$T_r = \frac{(V_{dr} + BR_c I_d)\pi}{3\sqrt{2}BE_{ac-rated}\cos(\alpha)} = 0.8541$$

[1 mark]

ii)

Reduced AC voltage at the rectifier end

$E_{ac-r}{}' = 0.9 \times 400$ kV $= 360$ kV

[1 mark]

Revised gate delay angle needed to maintain the direct current

$$\cos(\alpha) = \frac{(V_{dr} + BR_c I_d)\pi}{3\sqrt{2}BT_r E_{ac-r}{}'} > 1$$

[1 mark]

There is no firing angle that allows the rectifier to maintain the direct current. So the inverter takes over current control.

[1 mark]

Model Answers

The modified current reference

$$I_d{}' = I_d\left(1 - \frac{I_m}{100}\right) = 3.149\ \text{kA}$$

[1 mark]

At the rectifier end, the firing angle is stuck at the minimum value while the effective turns ratio is adjusted to maintain the rated direct voltage.

[1 mark]

$$T_r{}' = \frac{(V_{d-rated} + BR_c I_d{}')\pi}{3\sqrt{2}BE_{ac-r}{}'\cos(\alpha_{min})} = 0.8754$$

[1 mark]

Direct voltage at the inverter end

$$V_{di}{}' = V_{dr}{}' - R_L I_d{}' = 781.11\ \text{kV}$$

[1 mark]

At the inverter end, the effective turns ratio is adjusted to produce this direct voltage with minimum extinction angle.

$$T_i{}' = \frac{(V_{di}{}' + BR_c I_d{}')\pi}{3\sqrt{2}BE_{ac-rated}\cos(\gamma_{min})} = 0.8006$$

[1 mark]

iii)

Reduced AC voltage at the inverter end

$$E_{ac-i}{}' = 0.9 \times 400\ \text{kV} = 360\ \text{kV}$$

[1 mark]

Inverter end direct voltage

$$V_{di}{}'' = \frac{3\sqrt{2}}{\pi}BT_i E_{ac-i}{}'\cos(\gamma_{min}) - BR_c I_d = 714\ \text{kV}$$

[1 mark]

Rectifier end voltage

$$V_{dr}{}'' = V_{di}{}'' + R_L I_d = 736.49\ \text{kV}$$

Model Answers

[1 mark]

Maximum gate delay angle at the rectifier

$$\alpha = \cos^{-1}\left[\frac{(V_{dr}'' + BR_c I_d)\pi}{3\sqrt{2}BT_r E_{ac-rated}}\right] = 30.33°$$

[1 mark]

Model Answers

Question 3

a)

(i) The equations are as follows:

$$V_{DC}(t) - V_o(t) - V_{AC}(t) = R\,I(t) + L\,\frac{d}{dt}I(t)$$

[2 marks]

If this is transformed by Laplace, we obtain the following:

$$V_{DC}(s) - V_o(s) - V_{AC}(s) = R\,I(s) + L\,s\,I(s)$$

which gives the following transfer function between V_o and I:

$$\frac{I(s)}{V_o(s)} = \frac{-1}{Ls + R}$$

[2 marks]

(ii)

ii.a – The capacitance can be calculated backwards starting from the inertia time of the submodule:

$$H = \frac{E_{rated}}{P_{rated}} = \frac{\frac{1}{2}CV_{SM}^2}{P_{rated}}$$

This gives C=0.24 mF.

[2 mark]

ii.b – If the current of the converter is zero, there are no voltage drops across the filter impedances or the equivalent series resistance of the circuit. Therefore, the V_o^{DC}=500V and V_o^{AC}=230 V with an angle of 180º (ie in the opposite direction to V_{AC}).

[2 marks]

ii.c – We start by solving for the AC current required to deliver the requested power to the AC grid. Using the AC grid voltage as the reference for angles:

$$\bar{I}^{AC} = \frac{P_{rated}}{V_{AC}} = 13\text{ A}$$

Next we can calculate the AC voltage the stack must apply:
$\bar{V}_o^{AC} = 236.2\text{ V}$ with an angle of -172º.

[2 marks]

Knowing the voltage and the current of the stack, we can calculate the active power it is exchanging, which we will have to cancel out using the power exchanged using DC current:

$$P_o^{AC} = \Re e\left\{\bar{V}_o^{AC}\left(\bar{I}^{AC}\right)^*\right\} = -3{,}051\text{ W}$$

Here, candidates could ignore the resistance of the filter, which would have given an estimate of P_o of 3,000 W. This is perfectly reasonable as long as the assumption is states explicitly.
Next we can calculate the DC current knowing that the power to be taken from it is

$$P_o^{DC} = -P_o^{AC} = V_o^{DC} I^{DC}$$

If the resistance of the circuit is to be taken into account, this requires solving a system of two equations where two of the unknowns appear multiplied one by the other. Alternatively, if the DC resistance is neglected, we get:

Model Answers

$$V_o^{DC} = V_{DC} = 500 \text{ V}$$
$$I^{DC} = \frac{P_o^{DC}}{V_{DC}} = 6.1 \text{ A}$$

[3 marks]

(iii)

iii.a – The current references the controller has to track contain AC (50 Hz) and DC. The main disturbances affecting this current are of the same frequencies. This suggests that the controller has to have high gain at both frequencies. A PI controller has infinite gain at DC and a finite gain at 50 Hz that can be tuned by adjusting the bandwidth of the controller to make the closed-loop error at this frequency as small as required. Under these considerations and knowing that the error can't be made to be zero (but potentially small enough) with a PI, a PI is a reasonable choice.

[2 marks]

iii.b – The voltage at the output of the stack and the current going through the circuit contain AC (50 Hz) and DC. The instantaneous power exchanged by the bridges of the submodules and the submodules will contain the terms that correspond to the product between voltages and currents. Even without considering the limited frequency of switching of the converter, this means that the power going through the capacitor of the submodule contain 50 Hz and 100 Hz. This ripple is inevitable and it has to be ignored by the energy regulator. Typically this involves making the bandwidth of the controller low enough or by modifying the controller to include resonant notches at these frequencies.

[3 marks]

b)

(i)

i.a – Considering an operating point at zero power, there's no current flowing through the circuits, which makes the voltage drop of filters and effective series resistance be zero. This gives>

$V_{AC}^u = -V_{AC} = 400$ V with an angle of 180º (using the AC grid voltage as the reference of angles)

$V_{DC}^u = \frac{V_{DC}}{2} = 700$ V

$V_{AC}^l = V_{AC} = 400$ V with an angle of 0º.

$V_{DC}^l = \frac{V_{DC}}{2} = 700$ V

[2 marks]

i.b – The submodule must output a voltage that is at its greatest point equal to the DC component plus the peak value of the AC component. This gives: $\{V_u\}_{peak} = 700 + \sqrt{2}\, V_{AC}^u = 1.27$ kV This voltage is about twice as big as the submodule voltage (ie 700V). Therefore, each arm requires at least 2 submodules. This gives about 11% additional voltage margin for control headroom and to deal with variations of AC and DC grid voltage.

[3 marks]

Model Answers

(ii) The only way to make the power extracted from the positive pole of the DC grid be different from the power of the negative pole is to have a different current going through them. Specifically, we would need $I_{DC}^{u} = \frac{P_{DCu}}{V_{DC}/2} = 29$ A and $I_{DC}^{l} = \frac{P_{DCl}}{V_{DC}/2} = 14.5$ A.

[2 marks]

While it is possible to have such current circulating through the circuit, this is to be avoided because it causes the difference between the two currents to flow through the neutral on the AC side of the converter, in a real scenario this would lead to transformer saturation and other problems if not designed with this specific requirement.

[1 marks]

(iii) This operating condition requires summative AC current flowing within the converter. We've been told that the power delivered to the AC grid is zero, which suggests that this current will be the only current flowing through the converter (ie no AC grid current and no DC current present). This condition enables us to solve for the differential AC voltage and the DC voltages applied by the converter, leaving the summative AC voltage as the only unknown. If we assume the summative voltage to be very small compared to the differential voltage, we can solve for the summative AC current as:

$$\bar{I}_{sum}^{AC} = \frac{P_{u\to l}}{V_{AC}} = 25 \text{ A}$$

This means that the currents would be: $I_u^{DC} = 0$, $\bar{I}_u^{AC} = 25\ A$, $I_l^{DC} = 0$, $\bar{I}_l^{AC} = 25\ A$.

[2 marks]

We can also find the summative AC voltage to confirm the hypothesis that it is comparatively smaller than the AC grid voltage:

$\bar{V}_{sum}^{AC} = -2(R_{arm} + iX_{arm})\bar{I}_{sum}^{AC} = 125$ V with an angle of 90º.

$\bar{V}_u^{AC} = -\bar{V}_{diff}^{AC} + \frac{1}{2}\bar{V}_{sum}^{AC} = 405$ V with an angle of -171º, which is very similar to 400 V with an angle of -180º.

[2 marks]

Model Answers